

70-467 Machine Learning for Business Analytics

Class 2: Linear Regression

Professor: Andrés Castaño Zuluaga

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Outline

- 1 The linear model (simple regression)

- 2 Least squares

- 3 Accuracy of the coefficients and of the fit

- 4 Multiple regression

- 5 Qualitative predictors

Application: 400 stores, unit sales. We estimate the models in R.

Three uses of a regression

DESCRIPTIVE

How are the variables related?

EXPLANATORY

How does bad shelf location explain differences in sales?

PREDICTIVE

What will be the estimated sales for next quarter?

Explanatory means association after controls, not a causal effect. Prices, advertising, and shelves were not randomly assigned.

Estimating f

A quantitative response is written as

$$Y = f(X) + \varepsilon,$$

where f is unknown and ε is a mean-zero error, independent of X .

Two reasons to estimate f :

- **Prediction.** Produce $\hat{Y} = \hat{f}(X)$. The exact form of $\hat{f}$ can be secondary if $\hat{Y}$ is accurate.
- **Inference.** Understand how Y is associated with $X_1, \dots, X_p$. Then $\hat{f}$ cannot be a black box.

Linear regression does both. It is restrictive about the form of f , and that restriction is what makes the coefficients readable.

Regression, not classification

Regression: Y is quantitative. Here, unit sales.

Classification: Y is a category. Later in the term (logistic regression).

A linear model for a quantitative Y is

$$f(X) = \beta_0 + \beta_1 X_1 + \cdots + \beta_p X_p.$$

Simple linear regression is the case $p = 1$. Multiple linear regression is $p > 1$. The error term is still there:

$$Y = \beta_0 + \beta_1 X_1 + \cdots + \beta_p X_p + \varepsilon.$$

Simple linear regression

One predictor. The **population** regression line is

$$Y = \beta_0 + \beta_1 X + \varepsilon.$$

- β_0 : intercept — $E(Y \mid X = 0)$
- β_1 : slope — average change in Y associated with a one-unit change in X
- ε : everything the line misses (other variables, measurement error, nonlinearity)

We do not observe β_0 and β_1 . We observe n pairs (x_i, y_i) and compute estimates $\hat{\beta}_0, \hat{\beta}_1$. The **fitted** line is

$$\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i, \quad e_i = y_i - \hat{y}_i.$$

e_i is the i th **residual**.

The running example

A firm selling child car seats. File on **400 stores**: sales, own price, competitor price, local advertising, shelf quality, and neighborhood variables.

Headquarters is considering price, advertising, and placement.

Outcome Y : *Sales* — unit sales at the store, in thousands.

Predictor we start with: *Price* — list price, in dollars.

Variables in the file

Variable	Definition
USED IN TODAY'S MODELS	
Sales	Unit sales (thousands)
Price	Own list price (dollars)
CompPrice	Competitor's price (dollars)
Advertising	Local advertising budget (thousands of dollars)
ShelveLoc	Shelf quality: Bad, Medium, Good
Income	Community income (thousands of dollars)
IN THE FILE; NOT IN TODAY'S SPECIFICATIONS	
Population, Age, Education, Urban, US	Neighborhood and location indicators

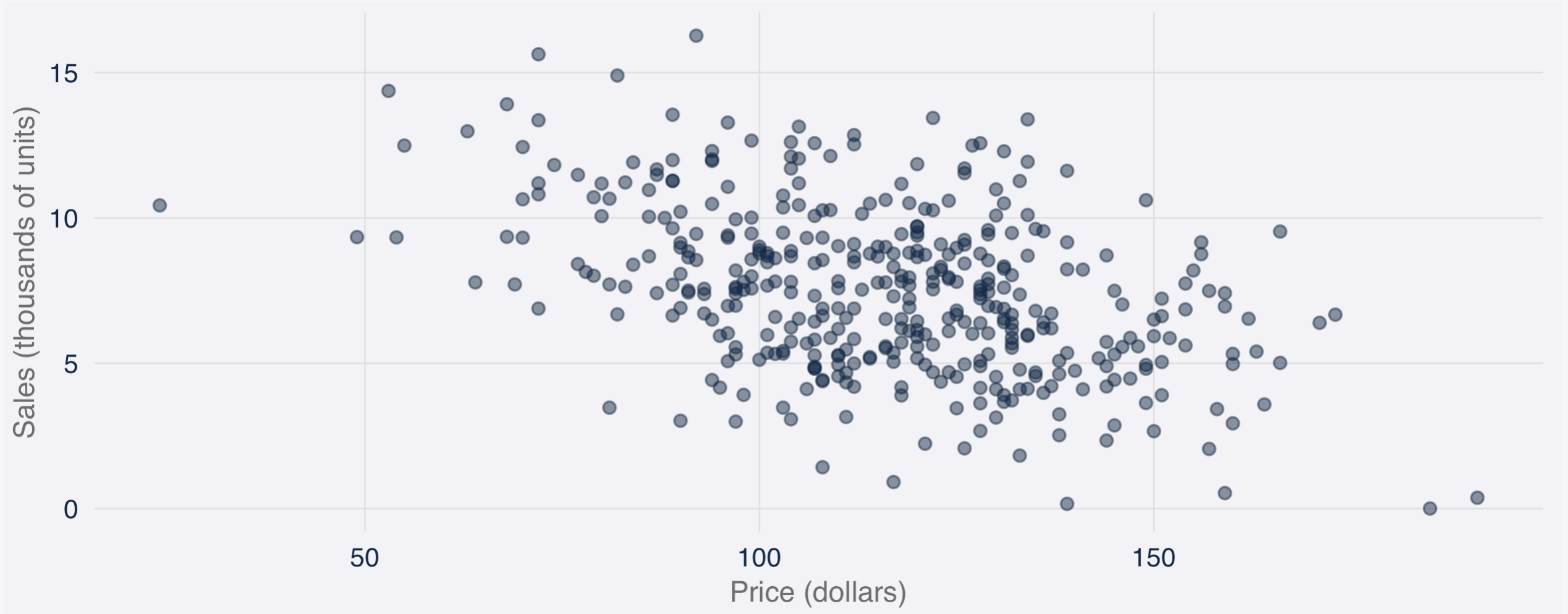
What we want from the model

Questions a linear model can answer in this file:

- 1 Is price associated with sales?
- 2 How strong is that association?
- 3 Which other predictors are associated with sales once price is in the model?
- 4 How large is each association, in units of sales?
- 5 How accurately can we predict sales for a store with given X ?
- 6 Is a linear specification reasonable?
- 7 Not today. Do predictors interact (synergy)?

Linear regression addresses (1)–(6). Interactions among predictors come later.

Sales against price



$n = 400$. Correlation of Price with CompPrice is 0.58 — relevant once we have more than one predictor.

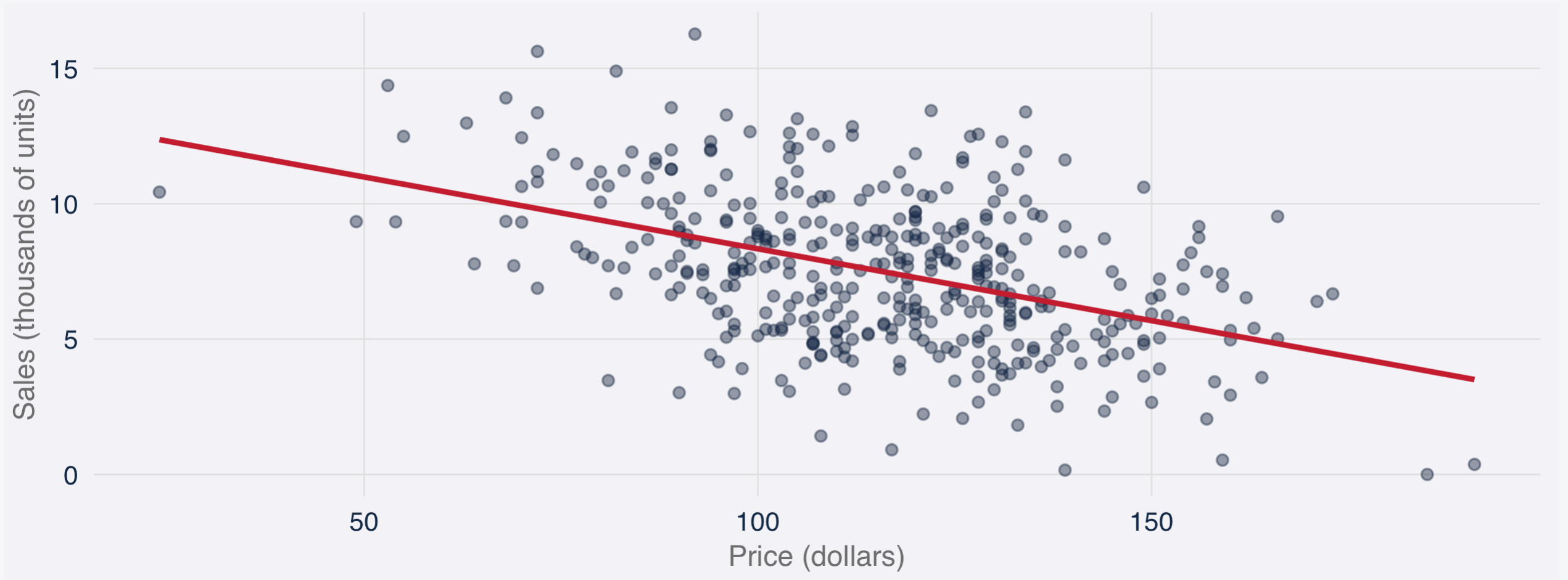
Least squares

Choose $\hat{\beta}_0$ and $\hat{\beta}_1$ to minimize the **residual sum of squares**

$$\text{RSS} = \sum_{i=1}^n (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i)^2 = \sum_{i=1}^n e_i^2.$$

The minimizers have a closed form. We compute them rather than derive them.

Least squares fit



Estimating the simple regression in R

```
1 m1 <- lm(Sales ~ Price, data = stores)
2 coef(m1)
```

```
(Intercept)    Price
13.64191518 -0.05307302
```

`Sales ~ Price` is the model formula: Sales as a function of Price, plus an intercept.

$RSS = 2,552.2$. $n - 2 = 398$ residual degrees of freedom.

Coefficient estimates

Term	Estimate	Std. Error	t	p-value
(Intercept)	13.6419	0.6328	21.56	< 0.00
Price	-0.0531	0.0054	-9.91	< 0.00

- **Estimate.** $\hat{\beta}$.
- **Std. Error.** Estimated standard deviation of $\hat{\beta}$ across samples of this size.
- **t.** $\hat{\beta}/SE(\hat{\beta})$. How many standard errors $\hat{\beta}$ is from zero.
- **p-value.** Probability, under $H_0: \beta = 0$, of a $|t|$ at least this large.

Interpretation of $\hat{\beta}_0$ and $\hat{\beta}_1$

Coefficient	Estimate	Meaning in this file
Intercept $\hat{\beta}_0$	13.64	Fitted sales if Price = 0. There is no such store. The intercept is not a business quantity here.
Price $\hat{\beta}_1$	-0.053	A one-dollar higher price is associated with -0.053 thousand fewer units — about 53 seats — in this simple regression.

A ten-dollar difference in price, using only this slope:

$$10 \times (-0.053) = -0.53$$

thousand units. That is an association in 400 observed stores. It is not the effect of an experimental price cut.

Standard errors and a confidence interval

If the errors are uncorrelated with constant variance σ^2 ,

$$\text{SE}(\hat{\beta}_1)^2 = \frac{\sigma^2}{\sum_{i=1}^n (x_i - \bar{x})^2}.$$

σ is unknown. We estimate it from the residuals (next slide). For Price,

$$\text{SE}(\hat{\beta}_1) = 0.0054.$$

An approximate 95% confidence interval is $\hat{\beta}_1 \pm 2 \text{SE}(\hat{\beta}_1)$:

$$[-0.064, -0.042].$$

The interval does not contain zero. That is the same information as the t of -9.9 on the previous table.

Testing $H_0: \beta_1 = 0$

$$H_0: \beta_1 = 0 \quad \text{versus} \quad H_a: \beta_1 \neq 0.$$

Under H_0 , there is no linear association between Price and Sales. The test statistic is

$$t = \frac{\hat{\beta}_1 - 0}{\text{SE}(\hat{\beta}_1)} = -9.91.$$

The p -value is < 0.001 . We reject H_0 .

A small p -value is not a recommendation to cut the price. It is evidence against “no linear association” in this specification, in this sample.

Residual standard error

Even if β_0 and β_1 were known, ε would remain. The **residual standard error** estimates σ :

$$\text{RSE} = \sqrt{\frac{1}{n-2} \sum_{i=1}^n e_i^2} = \sqrt{\frac{\text{RSS}}{n-2}}.$$

For this fit, $\text{RSE} = 2.53$ thousand units. Typical deviation of actual sales from the fitted line, in the units of Y .

$\text{RMSE} = \sqrt{n^{-1} \sum e_i^2}$ divides by n instead of $n - 2$. Here it is 2.53. We report **RSE**.

Mean sales in the file: 7.50 thousand. RSE is about 34% of that mean.

R^2

RSE is scale-dependent. R^2 is the fraction of variance in Y explained by the model:

$$R^2 = 1 - \frac{\text{RSS}}{\text{TSS}}, \quad \text{TSS} = \sum_{i=1}^n (y_i - \bar{y})^2.$$

TSS is the RSS of the intercept-only model (predict $\bar{y}$ for every store).

For Sales on Price, $R^2 = 0.198$. About one fifth of the variation in sales is associated with price in this linear fit.

In applications of this kind, R^2 well below 1 is typical. $R^2 = 0.20$ is not, by itself, a failed model. It is also not a reason to stop at one predictor.

Why a single predictor is not the specification we want

Price and CompPrice move together. A store that charges more often sits next to a competitor that charges more.

If we omit CompPrice, part of what we call “the price coefficient” is the local market.

Multiple linear regression includes several predictors at once. The coefficient on Price is then the association with Sales **holding the other included predictors fixed**.

Multiple linear regression

$$Y = \beta_0 + \beta_1 X_1 + \cdots + \beta_p X_p + \varepsilon,$$

or, equivalently,

$$\mathbf{E}(Y \mid X_1, \dots, X_p) = \beta_0 + \beta_1 X_1 + \cdots + \beta_p X_p.$$

β_j is the average change in Y associated with a one-unit change in X_j , **holding** $X_1, \dots, X_{j-1}, X_{j+1}, \dots, X_p$ **fixed**.

Least squares still minimizes RSS. The geometry is a hyperplane instead of a line. The algebra is the same idea.

Two predictors that belong together

```
1 m2 <- lm(Sales ~ Price + CompPrice + Income, data = stores)
```

Term	Estimate	Std. Error	t	p-value
(Intercept)	4.9502	0.9814	5.04	< 0.00
Price	-0.0872	0.0058	-14.99	< 0.00
CompPrice	0.0928	0.0090	10.31	< 0.00
Income	0.0153	0.0040	3.81	< 0.00

$R^2 = 0.381$. RSE = 2.23.

The Price coefficient is not invariant to the specification

Model	$\hat{\beta}_{\text{Price}}$	R^2	RSE
Sales ~ Price	-0.053	0.198	2.53
+ CompPrice + Income	-0.087	0.381	2.23

After CompPrice and Income are included, a one-dollar higher own price is associated with a **larger** sales gap (-0.053 versus -0.087).

$\hat{\beta}_{\text{Price}}$ is a different number once the competitor's price is held fixed. CompPrice itself is 0.093: holding own price and income fixed, a one-dollar higher competitor price is associated with about 93 extra seats.

Advertising

```
1 m3 <- lm(Sales ~ Price + CompPrice + Income + Advertising, data = stores)
```

$\hat{\beta}_{\text{Advertising}} = 0.131$. Holding the other three predictors fixed, an additional 1,000 dollars of local advertising is associated with 0.131 thousand extra units.

R^2 moves from 0.381 to 0.475. The Price coefficient remains about -0.091.

The table does not contain margin or the cost of the advertising. The coefficient is a sales association, not a return on spend.

Qualitative predictors

`ShelveLoc` takes three values: Bad, Medium, Good. It is not a number.

For a qualitative predictor with k levels, the linear model uses $k - 1$ **dummy variables**. One level is the baseline.

Code `ShelveLoc` so that Bad is the baseline. Then

$$\begin{aligned} E(\text{Sales} \mid \dots) = & \beta_0 + \beta_1 \text{Price} + \dots \\ & + \beta_{\text{Med}} \mathbf{1}\{\text{Medium}\} + \beta_{\text{Good}} \mathbf{1}\{\text{Good}\}. \end{aligned}$$

- β_{Good} : difference in mean sales, Good versus Bad, **holding the other included predictors fixed**
- β_{Med} : the same comparison for Medium versus Bad

Model with shelf location

```
1 m4 <- lm(Sales ~ Price + CompPrice + Income + Advertising + ShelfLoc,  
2          data = stores)
```

Term	Estimate	Std. Error	t	p-value
(Intercept)	2.4313	0.5690	4.27	< 0.00
Price	-0.0932	0.0033	-28.24	< 0.00
CompPrice	0.0957	0.0051	18.76	< 0.00
Income	0.0160	0.0023	7.05	< 0.00
Advertising	0.1162	0.0096	12.15	< 0.00
ShelveLocMedium	1.8499	0.1550	11.93	< 0.00
ShelveLocGood	4.7977	0.1888	25.41	< 0.00

$R^2 = 0.803$. RSE = 1.26.

Reading the dummy coefficients

Holding Price, CompPrice, Income, and Advertising fixed:

- Medium versus Bad: 1.85 thousand units
- Good versus Bad: 4.80 thousand units

A ten-dollar own-price difference in this specification is about $(10 \cdot -0.093)$ thousand units.

Shelf quality is the large association in this model. That does not imply “ignore price.” It implies that any recommendation needs to consider the **cost** of moving each lever. Cost is not in the file.

Two stores that differ **only** in shelf (Good vs Bad), with the other included X the same, differ in fitted sales by $\hat{\beta}_{\text{Good}}$. That is what “holding the other predictors fixed” means.

Nested specifications

Model	Price	R ²	RSE
Sales ~ Price	-0.053	0.198	2.5
+ CompPrice + Income	-0.087	0.381	2.2
+ Advertising	-0.091	0.475	2.0
+ ShelveLoc	-0.093	0.803	1.2

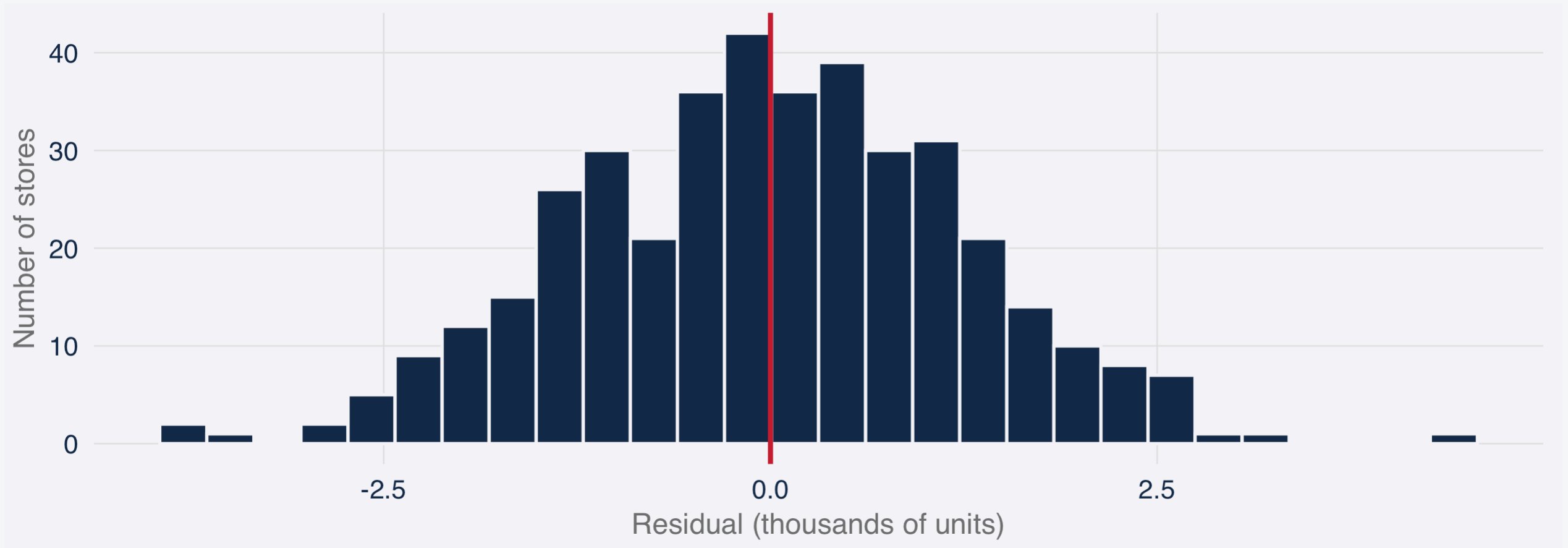
When a predictor is added, two questions:

- 1. Does the coefficient you care about **change**?
- 2. Does R^2 rise, and RSE fall, enough to matter for the decision?

R^2 always rises (weakly) when a predictor is added. That is not, by itself, a reason to include it. Adjusted R^2 and an F test compare nested models without that mechanical increase.

Residuals from the last model

$e_i = y_i - \hat{y}_i$. RSE = 1.26 thousand units (simple regression: 2.53).



What the estimates are not

Not an experiment. Prices, ads, and shelves were not randomly assigned. $\hat{\beta}_{\text{Price}}$ is not “the effect of cutting the price.”

Not a prediction for new stores. Every R^2 and RSE here uses the same 400 rows used to fit. In-sample fit overstates accuracy out of sample.

Not immune to collinearity. Price and CompPrice are correlated. If they were nearly the same series, the individual coefficients would be unstable even if fitted values remained usable.

What you should be able to do

- Write the simple and multiple linear models, including ε .
- Obtain $\hat{\beta}$ by least squares in R and read Estimate, SE, t , and p .
- Interpret $\hat{\beta}_j$ **in units**, holding other included predictors fixed.
- Report RSE (units of Y) and R^2 (share of variance), and know they answer different questions.
- Explain why $\hat{\beta}_{\text{Price}}$ changed when CompPrice entered.
- Encode a k -level factor as $k - 1$ dummies and interpret the coefficients against the baseline.
- State what you will **not** claim from an observational fit scored in-sample.

Next class

Monday 31 August. Quiz 1 at the start (paper, 10–15 minutes, closed notes). Then multiple regression in more depth: the overall F test, nested comparisons, and qualitative predictors.

Wednesday 2 September. First lab, a different data set. Homework 1 goes out that day, on a third case.

If R or RStudio is still not working, see me after class today.

Before Monday

James, Witten, Hastie, Tibshirani, *An Introduction to Statistical Learning*, 2nd ed. Free at statlearning.com.

Chapter 3, pages 71–82

Multiple linear regression, and the questions one asks of a model with several predictors.

Chapter 3, pages 83–86

Qualitative predictors. Stop before page 87.

Pages 63–68, if you want another pass

Standard errors, confidence intervals, and t-statistics.